MATTHEW H. TAYLOR

mht3.github.io

Research Focus: AI for scientific discovery, physical applications of deep learning and reinforcement learning.

EDUCATION

UNIVERSITY OF CALIFORNIA, SAN DIEGO — **SEP. 2023 – JUNE 2025**
- MS in Computer Science and Engineering — *GPA: 4.0 / 4.0*

UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN — **AUG. 2019 - MAY 2023**
- Bachelor of Science in Aerospace Engineering — *GPA: 3.93 / 4.00*
- Highest Honors, Minor in Computer Science

SKILLS

- Python
- C/C++
- Java
- PyTorch
- Git
- Unix
- Scikit-learn
- TensorFlow

WORK EXPERIENCE

NASA Jet Propulsion Laboratory, *Machine Learning Research Intern* — **JUNE 2024 - NOV. 2024**
- Leveraged self-supervised contrastive learning techniques to create an ocean synthetic-aperture radar (SAR) foundation model and finetuned on ocean science object detection and semantic segmentation tasks.
- Developed a versatile, dataset and model-agnostic semantic segmentation and object detection training codebase with integration of Weights and Biases for visualizing and monitoring model performance.
- Curated a dataset with bounding box and polygon mask annotations for over 7,000 submesoscale ocean eddies in the western Mediterranean Sea and *submitted our work to CVPR 2025 as first author* (under review).

Dolby Laboratories*, Computer Vision/ML Engineer Intern* — **JUNE 2023 - AUG. 2023**
- Profiled video content using skin segmentation, noise, face, and text detection models to give suggestions for when to use precision detail, a local image adjustment algorithm created for Dolby Vision.
- Updated Dolby Vision's internal C++ to Python bridge by adding hooks for local enhancement parameters.
- Produced a modular PyQt GUI to visualize content mapped Dolby Vision content combined with model outputs. The modularity allows for new models to be used if a faster or more accurate algorithm is found.

Maxar Technologies*, Research and Development Intern* — **MAY 2022 - AUG. 2022**
- Applied deep learning and semantic segmentation methods to detect building footprints from Maxar satellite imagery using PyTorch Lightning for training and TensorBoard for model evaluation and visualizations.
- Researched how the addition of the near infra-red band (NIR) improves segmentation performance.
- Developed model-agnostic visualizations in TensorBoard for the R&D Lab's custom training framework.

RESEARCH AND TEACHING EXPERIENCE

Graduate Researcher*, Advised by Professor Sicun Gao* — **MARCH 2024 - PRESENT**
- Improving learning-based design for nuclear fusion plasma control using Lyapunov certificates of stability.
- Created an approximate neural Lyapunov controller for pendulum and cartpole gymnasium environments with the goal of expanding and generalizing to problems with larger state spaces and unknown dynamics.

Teaching Assistant, *Intro to Programming (CSE 8A)* — **JAN. 2024 - MARCH 2024**
- Led discussion sections and held office hours for a class of nearly 300 students.
- Created quizzes and homework assignments to enrich student learning objectives.

Undergraduate Researcher*, Tran Research Group* — **MAY 2021 - MAY 2023**
- Created a CNN to detect objects and transform a robot's camera view into a top-down grid world in simulation and the real-world using the NVIDIA JetBot AI kit. Controlled the robot in simulation using Gazebo and ROS.
- Explored useful applications for brain computer interfaces and human-agent teaming, applying machine learning algorithms to EEG data to improve a human-agent collaborative environment.
- Created an experiment with multiple participants and developed a multiclass SVM to classify LEDs flashing at different frequencies based on EEG signals using steady-state visual-evoked potentials (SSVEP).

RELEVANT COUSEWORK

- Search and Optimization
- Deep Learning - 3D Data
- Introduction to AI
- Algorithms
- Probability and Statistics
- Introduction to Robotics
- ML for Physical Appl.
- Parallel Computing
- Machine Learning